Rule-based machine learning

Apriori

Market Basket Analysis

- Transaction Data
  - Forget the sample, it's the feature
  - Binary

Association Rules

- Causality (direct dependency) A→B
- Correlation (strength & direction of linear relation)

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- Causality (direct dependency) A→B
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- Association (gives info about, any relation)
  - May have a common cause
    - Coffee prevents skin cancer?
  - Behavior modeling
  - Frequency and co-occurrence

Finding association rules

- An itemset
- A frequent itemset
- An association rule
  - body (antecedent) AND head (consequent)
  - {Bread, Butter} => {Milk}
- Many potential rules within one frequent itemset
  - Rule: split and imply

Frequent itemsets are expensive

- Based on num. unique features
  - Exponential growth of possible itemsets
  - Hyperparams to control

Apriori

- Freq. itemsets are anti-monotonic in set theory

All non-empty subsets of a frequent itemset must be frequent.

=

If an itemset is infrequent, all its supersets will be infrequent.

- Reduce search space in candidate generation

Terminology

- *Transaction:* sample
- *Item/Itemset/k_itemset*: feature/set of features/ set of k features

- Evaluation & Thresholding
  - ***Itemset Support*** *(itemset)* : How often does itemset happen?

    trans. itemset / total trans.

  - ***Rule Confidence*** *(body→head)*: When *head*, how often *body*?

    trans. *body* AND *head* / trans. body

Terminology II

- ***Rule Lift*** *(body→head): dependence between prob. of body and head*

$$\frac{support(body \rightarrow head)}{support(body) * support(head)} = \frac{confidence(body \rightarrow head)}{support(head)}$$

- Lift = 1 : Independence between *body* and *head.*
- Lift < 1 : *body* has inhibition effect on *head*
- Lift > 1 : *body* has presence effect on *head*

Pruning

- *Support* to reduce search space of itemsets
- Confidence to discard irrelevant rules
- *Of special interest:*
  - *Maximal itemset*: No superset is freq. itemset
  - *Closed itemset*: All supersets have lower support

Apriori in action

```
itemsets = all_features

For K=1…N

    freq_itemsets += [i in itemsets of size k & above support]

    itemsets= pair_wise_union(freq_itemsets)
```

Limitations

- Transaction data
  - Binary only!
  - Samples are clones
    - Pre-clustering & merging to mitigate

Limitations

- Num. candidates to check explodes with
  - Many itemsets
  - Large itemsets
  - Low support

- FP-Growth
  - One order of magnitude faster. Still limited.

By hand

- freq_itemsets = all itemsets
- filter freq_itemsets by support
- k = 2
  - Combinate freq_itemsets to make larger itemsets (k+1)
    - pairs sharing k-1 elements
  - filter freq_itemsets by support
  - if empty, break
  - produce association rules

Bicing	***Metro***	***Bus***	***Moto***	***Cotxe***
X		X		
	X			X
X	X		X	
X	X	X		
	X		X	
	X	X	X	
X	X			

- Min. itemset support: 2